Mohammad Ali Asif - ECE - 2nd year - Probability assignment

Computer A sends a message to computer B over an unreliable radio link.

The message is encoded so that B can detect when errors have been introduced

into the message during transmission. If B detects an error, it requests A to retransmit it.

If the probability of a message transmission error is q = 0.1.

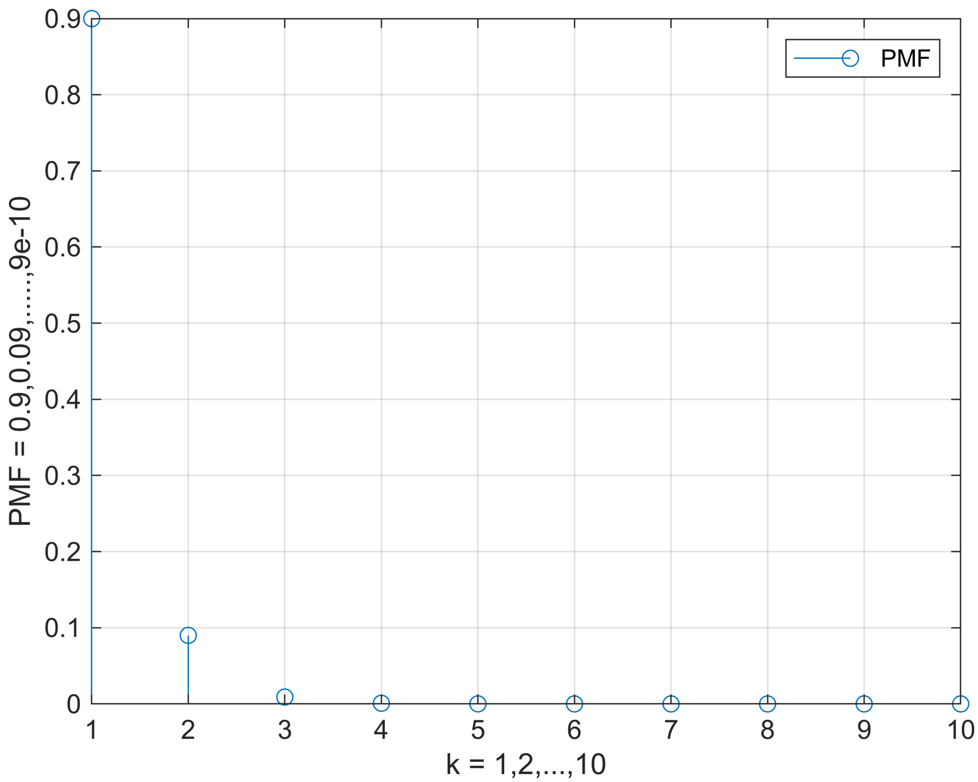
1. Plot the PMF of the resulting geometric random variable.

Recall the second version of geometric random variable rule, where the number of trails is noted.

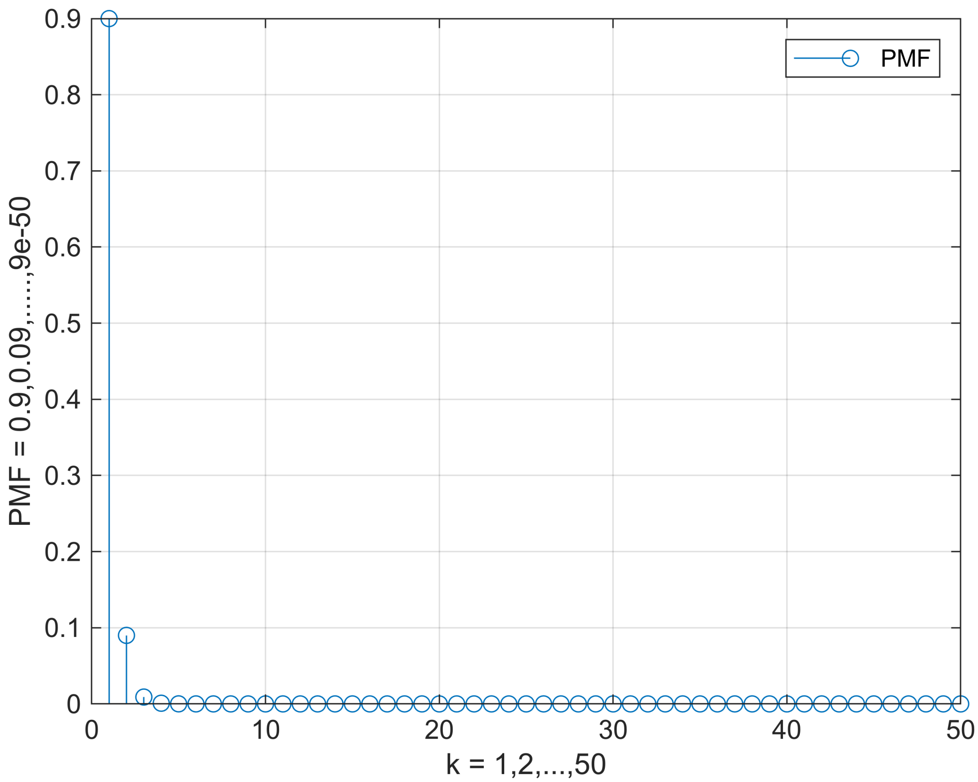
pmf = p*(1-p)^k = p*q^k

k = 1,2,.......... where k keeps increasing to infinity.

```
clear;clc;close all;
q = 0.1;
p = 1-q;
k = 1:10;
pmf = p*(q.^(k-1));
stem(k,pmf)
grid on
legend('PMF')
xlabel('k = {1,2,...,10}')
ylabel('PMF = {0.9,0.09,.....,9e-10}')
```




```
k = 1:50;
pmf = p*(q.^(k-1));
stem(k,pmf)
grid on
legend('PMF')
xlabel('k = {1,2,...,50}')
ylabel('PMF = {0.9,0.09,.....,9e-50}')
```



so in this case where p = 0.9, we are always going to get Pk = 9 x 10^-k.

2.What is the probability that a message needs to be transmitted more than two times?

we have q = 0.1, P(k > 2) = q^k = 0.1^2 = 0.01

```
p2 = q^2
```

```
p2 = 0.0100
```

or we could do: 1 - summation(PMF(k<=2)), that would work too.

```
p2 =  1 - sum(pmf(k<=2),2) % where sum(V,2) is the sum of the V elements in each row.
```

```
p2 = 0.0100
```

The probability of retransmitting the message over 2 times is too low

which makes it almost impossible to happen, we may notice that using

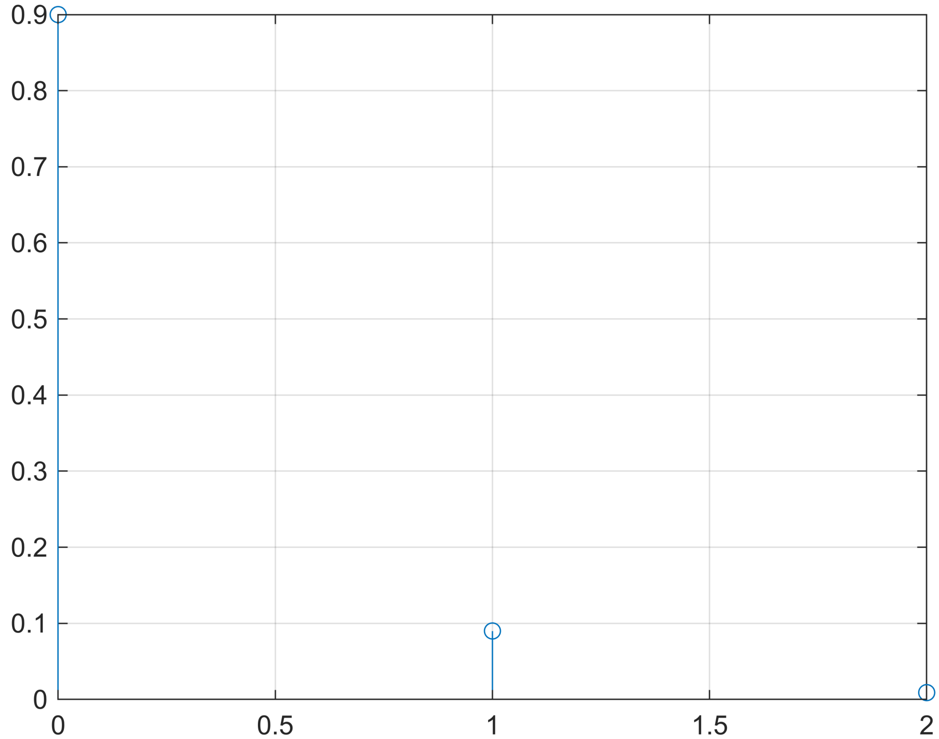
geornd function that generates random numbers that represent the failures in sending the message.

which means k = 0,1,.......... in this case.

```
k = geornd(p,[1 200]) % geornd(p, size of array/matrix of generated random numbers)
```

```
k = 1×200
     0     0     0     0     0     0     0     0     1     0     0     0     1···
```

```
pmf = p*(q.^k);
% notice that we used the first version of geometric random variables rule
% because the function counts the number of failures, and
% not trails as the second version.
stem(k,pmf); grid on;
```



200 random generated numbers but it barely could reach 2 failed sent messages with the error probability q = 0.1